

## Supplementary Methods and Results

*Companion to: Consumed Methadone Dose Patterns Around Randomly Scheduled Urine Testing: Pre-Test Escalation and Higher-Risk Trajectories in Long-Retained Maintenance Patients*

These analyses are referenced from the Methods and from Sensitivity analyses in the main text. They are reported in full here so that the main text can state conclusions without reproducing every intermediate result.

### S1. Additional sensitivity analyses

**Independence from attendance.** Because consumed dose could be confounded with attendance (the dominant single association), each cluster's association was re-estimated adjusting for the pre-test attendance rate. All nine clusters remained significant (FDR  $q < 0.05$ ), but the adjustment is not uniformly neutral: 7d-1 (OR 1.75→1.67), 7d-4 (1.27→1.27), 7d-5 (0.47→0.50) and PCA-3 (1.37→1.42) are essentially unchanged, whereas **7d-6 attenuates substantially (1.62→1.29) and 7d-2 attenuates (1.67→1.37)**. Attendance therefore explains part, though not all, of two of the four higher-risk dose-change associations.

To exclude the stronger possibility that the dose-change clusters are an artefact of *missing* dosing days — a cluster shaped by absence, and by the interpolation used to fill it (Inclusion and feature construction in the main text) — the analysis was repeated on the subset with **complete attendance in the pre-test week**: the 6,631 tests (67%; 1,816 positive, 4,815 negative) with an attendance proportion of 1.0 over days  $-7 \dots -1$ .

**This subset is not imputation-free over  $\pm 7$  days, and earlier versions of this manuscript wrongly described it as such. The subsetting variable constrains only the pre-test week (which does, as it happens, guarantee an observed day-(-7) normalisation anchor: 0 of the 6,631 have an imputed anchor), but it leaves days 0 ... +7 unconstrained, and 987 of the**

6,631 tests (15%) contain at least one interpolated day inside  $\pm 7$ . The genuinely imputation-free subsets — built directly from the pre-fill series, in which no day inside the stated window carries the missing-day code — are smaller: 5,644 tests (57%; 1,451 positive, 4,193 negative) over  $\pm 7$ , and 3,737 (38%) over  $\pm 28$ . Every imputation claim under Sensitivity analyses in the main text is computed on those (`42_correct_imputation_free.py`); the cluster-level figures below are retained because they are correct for the subset actually used, now correctly named. Reporting all nine clusters, and naming the two that change label: 7d-1 P/N 1.95, 7d-2 1.45, 7d-4 1.43 and PCA-3 1.44 remain higher-risk ( $q < 0.05$ ); 7d-3 0.52, 7d-5 0.51 and PCA-2 0.68 remain lower-risk ( $q < 0.05$ ); but 7d-6 falls from P/N 1.52 to 1.20, landing on rather than above the 1.20 higher-risk threshold ( $q = 0.04$ , so still significant but no longer labelled higher-risk), and PCA-1 rises from 0.77 to 0.85, just above the 0.83 lower-risk threshold ( $q = 0.02$ , likewise significant but no longer labelled). The dose-change partition therefore survives restriction to complete pre-test-week attendance for three of its four higher-risk clusters and both protective clusters, but 7d-6 — the cluster that looked strongest against observed-collection negatives — is roughly halved on the log scale, consistent with the attendance-adjusted result above. (Because this subset is not imputation-free, these figures bound the influence of non-attendance, not of the interpolation; the latter is isolated under Sensitivity analyses in the main text.) The stable-dose branch is not protected against the imputation. The subset above constrains only the pre-test week, whereas the stable-dose clusters are built from  $\pm 28$ -day series that remain largely interpolated within it. Restricting instead to the 3,737 tests (805 positive, 2,932 negative) with a fully observed  $\pm 28$ -day window, all three stable-dose clusters retain their direction but none remains significant (PCA-1 P/N 0.84, PCA-2 0.78, PCA-3 1.39; all  $q \approx 0.07$ –0.14). Given

the smaller sample this is not evidence of absence, but it does mean the stable-dose clusters have no complete-data corroboration — a further reason (with Clustering in the main text) to treat that branch as exploratory.

**The partition adds over a single scalar.** To test whether the nine clusters merely discretise one continuous escalation slope, we compared logistic models. Cluster membership explained substantially more variance than a continuous post-test slope (McFadden pseudo- $R^2$  0.028 vs 0.008), and adding the clusters to the scalar improved fit decisively (likelihood-ratio  $\chi^2 = 261$ ,  $df = 8$ ,  $p < 10^{-50}$ ). The shape information captured by the partition is therefore not reducible to a single dose-change magnitude.

**The higher-/lower-risk split is not an artefact of the chosen number of clusters; the K-sweep does not, however, establish the ordering.** Because  $K = 6$  was our choice rather than a criterion's (Clustering in the main text), the dose-change branch was refitted at  $K = 4, 5, 6, 7$  and 8 and the cluster–outcome pattern re-estimated within that branch (39\_k\_robustness.py; Table S4; models fitted within the dose-change branch with FDR across the clusters of each solution, so counts differ slightly from Table 2, which pools both branches). At every resolution the same structure appears — a set of higher-risk and a set of lower-risk clusters spanning a similar range of P/N ( $K = 4$ : 0.54–1.49, 2 higher / 2 lower;  $K = 5$ : 0.54–1.63, 2/2;  $K = 6$ : 0.50–1.64, 3/2;  $K = 7$ : 0.51–1.80, 4/2;  $K = 8$ : 0.51–1.73, 5/2).

Earlier versions of this manuscript also reported the rank correlation between a component's mean **post-test** escalation slope and its P/N as evidence for the *ordering*, quoting the  $\rho$  values without their p-values. Reported in full, that statistic does not support the claim, because it is computed on only  $K$  points:

| K (= n points) | $\rho$ , post-test ordering, full sample | p | $\rho$ , post-test, imputation-free $\pm 7$ | p | $\rho$ , pre-test ordering, imputation- | p |
|----------------|------------------------------------------|---|---------------------------------------------|---|-----------------------------------------|---|
|----------------|------------------------------------------|---|---------------------------------------------|---|-----------------------------------------|---|

|   |         |         |       |       | free $\pm 7$ |         |
|---|---------|---------|-------|-------|--------------|---------|
| 4 | +1.00 † | 0.083 † | +0.60 | 0.40  | +1.00 †      | 0.083 † |
| 5 | +0.80   | 0.104   | +0.30 | 0.62  | +0.60        | 0.285   |
| 6 | +0.71   | 0.111   | +0.43 | 0.40  | +0.77        | 0.072   |
| 7 | +0.86   | 0.014   | +0.68 | 0.094 | +0.64        | 0.119   |
| 8 | +0.57   | 0.139   | +0.33 | 0.42  | +0.90        | 0.002   |

69

70 † With  $n = 4$  and  $\rho = +1.00$  the asymptotic t-statistic is infinite and the software returns  $p = 0$ ; the  
71 value given is the **exact two-sided permutation p (1/12 = 0.083)**. This row is degenerate and  
72 carries no evidential weight.

73 Three things follow. First, only one of the five full-sample p-values ( $K = 7$ ) is nominally  
74 significant, and at  **$K = 6$  — the resolution used for Table 2 and every claim in this paper —  $p$**   
75  **$= 0.11$** . Second, the post-test ordering weakens further on the imputation-free subset and is non-  
76 significant at every  $K$ , which is consistent with the post-test *magnitude* being imputation-  
77 generated (Sensitivity analyses in the main text) and is a further reason not to claim it. Third, the  
78 ordering variable the paper does claim — the **pre-test** slope — is the one that behaves better  
79 once the imputation is removed, but a 4-to-8-point rank correlation cannot settle the question  
80 either way. We therefore withdraw the earlier statement that the ordering "is a property of the  
81 data, not of the resolution": what the K-sweep shows is that a higher-risk/lower-risk separation  
82 exists at every resolution. The ordering claim rests instead on the bootstrap gradient statistic  
83 reported under The dose-change partition and its ordering in the main text, which is estimated  
84 from 500 refits rather than from  $K$  points.

85 **The bootstrap gradient statistic (Table S7)**. For each of the 500  $K = 6$  bootstrap refits already  
86 used for Table S3, we recomputed the rank correlation between a component's mean escalation  
87 and its P/N, and the P/N spread between the top- and bottom-escalation components

(44\_gradient\_bootstrap.py). This is the statistic that distinguishes "cluster boundaries unstable" from "gradient unstable", and it is estimated from 500 replicates rather than from 4–8 points:

| Ordering variable       | Subset                  | Observed $\rho$ | Bootstrap $\rho$ , median (95%) | $P(\rho > 0)$ | Bootstrap top/bottom P/N spread, median (95%) | $P(\text{spread} > 1)$ |
|-------------------------|-------------------------|-----------------|---------------------------------|---------------|-----------------------------------------------|------------------------|
| pre-test (claimed)      | imputation-free $\pm 7$ | +0.77           | +0.77 (+0.49 to +0.94)          | 1.000         | 3.27 (1.18 to 4.95)                           | 1.000                  |
| pre-test (claimed)      | full, interpolated      | +0.60           | +0.49 (−0.14 to +0.71)          | 0.934         | 2.41 (0.57 to 3.43)                           | 0.852                  |
| post-test (not claimed) | imputation-free $\pm 7$ | +0.43           | +0.60 (+0.09 to +0.86)          | 0.990         | 2.62 (2.20 to 5.74)                           | 0.988                  |
| post-test (not claimed) | full, interpolated      | +0.71           | +0.89 (+0.57 to +1.00)          | 0.998         | 2.93 (1.20 to 7.21)                           | 0.996                  |

The row the paper relies on is the first: ordered by the component it claims, on windows containing no interpolated day, the gradient is positive in every one of 500 refits and both its rank correlation and its P/N spread exclude the null — while, on the same refits, no individual cluster's P/N interval excludes unity (The dose-change partition and its ordering in the main text). The second row is the honest counterweight: on the **full interpolated sample** the same statistic includes zero, so the gradient is established on imputation-free windows and not on the data as a whole. The post-test rows are reported for completeness only; that variable's magnitude is imputation-generated (Sensitivity analyses in the main text) and no claim is placed on it.

**Repeat same-day records: a conservative duplication check.** The pipeline retained same-patient-same-day repeat urine records (Cohort and participant flow in the main text). Test dates are not carried into the analysis file, but a same-day repeat necessarily produces a bit-identical 57-day raw dose window for the same patient with the same result, so those rows can be identified from the analysis file itself. That criterion is deliberately **over-inclusive**: it also

104 removes tests on genuinely different dates for patients whose dose was flat across both windows.  
 105 It flags 807 rows (88 positive, 719 negative) — a strict superset of the same-day repeats —  
 106 leaving 9,093 tests (3,313 positive, 5,780 negative). Recomputing the primary table on that  
 107 subset (45\_complement\_and\_dedup.py), **every cluster remains significant in the same**  
 108 **direction** (largest  $q = 0.012$ ) and the ranking is essentially preserved, with uniform mild  
 109 attenuation: 7d-2  $1.62 \rightarrow 1.54$ , 7d-1  $1.64 \rightarrow 1.53$ , 7d-6  $1.52 \rightarrow 1.42$ , PCA-3  $1.34 \rightarrow 1.26$ , 7d-4  
 110  $1.23 \rightarrow \mathbf{1.17}$ , PCA-1  $0.77 \rightarrow \mathbf{0.86}$ , PCA-2  $0.65 \rightarrow 0.62$ , 7d-3  $0.57 \rightarrow 0.55$ , 7d-5  $0.50 \rightarrow 0.47$ .  
 111 Two clusters cross a labelling threshold under this deliberately harsh check (7d-4 falls just below  
 112 1.2; PCA-1 rises just above 0.83), which is a further reason to read the thresholds as descriptive  
 113 conventions rather than as findings. Since the true duplicate count inside the analysable sample is  
 114 far smaller than 807, the real effect of retaining the repeats is smaller than this bound.  
 115 **Pre- versus post-test decomposition.** Separating the slopes, both the pre-test (OR 1.14 per SD,  
 116  $p = 0.01$ ) and the post-test (OR 1.28 per SD,  $p = 3 \times 10^{-5}$ ) escalation were associated with a  
 117 positive result, the post-test more strongly. A genuine pre-test behavioural component (rising  
 118 consumed dose *before* the result) therefore coexists with a stronger post-test component that  
 119 includes the clinic's dose response to a positive test — so the pattern is partly antecedent  
 120 behaviour and partly clinical reaction, and cannot be interpreted as purely predictive. This  
 121 ordering reverses once the imputation is removed: on imputation-free  $\pm 7$ -day windows the pre-  
 122 test component strengthens while the post-test component disappears (Sensitivity analyses in the  
 123 main text). The post-test magnitude reported here should therefore be read as an upper bound  
 124 obtained on the filled series, not as a robust estimate.  
 125 **Temporal validation.** This analysis uses the independent raw-data reconstruction (11,110 tests;  
 126 see the pipeline comparison below), so its attendance variable is rebuilt from observed dosing

days and has a smaller spread ( $SD \approx 0.15$ ) than the curated one; estimates are given on the same per-0.1 scale for comparability. Splitting the cohort at its median date (April 2020) and fitting each half independently, the attendance association replicated strongly in the held-out later period (**OR 0.86  $\rightarrow$  0.78 per 0.1 increment**, equivalently  $0.80 \rightarrow 0.68$  per within-period SD; both  $p < 10^{-7}$ ), as did prior testing history ( $OR \approx 2.8$  per SD). The weaker consumed-dose escalation association, however, **did not replicate out-of-time**: it was significant in the earlier period ( $OR\ 1.18$ ,  $p = 0.01$ ) but not in the later period ( $OR\ 1.15$ , 95% CI crossing 1,  $p = 0.17$ ) — so the dose effect, while directionally stable, is modest and not robust across time. Notably, the median split falls at the onset of the COVID-19 pandemic (early 2020), which disrupted clinic attendance and dispensing practice (e.g. take-home dosing) in MMT programmes generally; the pre/post-2020 split therefore doubles as a pandemic-era robustness check, and the persistence of the attendance and history associations across it argues against a purely pandemic-driven artefact, while the non-replication of the dose effect may partly reflect these operational changes rather than a change in patient behaviour.

**Confirmed observed-collection negative subset.** Because negatives collected under routine, unobserved conditions may be false negatives that dilute associations, we repeated the cluster analysis using only the higher-specificity negatives from patients with supervised urine collection: 1,210 negative tests (18.6% of negatives) from **63 patients**. This comparison requires care. Comparing *all* 3,401 positives against these 1,210 negatives compares two different populations: only 327 positives (9.6%) come from the supervised patients, and the supervised patients are systematically lower-risk (mean per-patient positive rate 0.197 vs 0.389 in the remaining 489 patients). We therefore report both the original comparison and a **like-for-like**

comparison restricting positives to the same 63 supervised patients (327 positives vs 1,210 negatives). Full results for all nine clusters are in Table S1 and Figure 7.

**The pattern is mixed, not uniformly strengthened.** Against the observed-collection negatives, three of the four higher-risk dose-change clusters **attenuate**: 7d-1 P/N  $1.64 \rightarrow 1.38 \rightarrow 1.31$  (primary  $\rightarrow$  all-positives  $\rightarrow$  like-for-like), 7d-2  $1.62 \rightarrow 1.24 \rightarrow 1.18$  (below the paper's own 1.2 higher-risk threshold), 7d-4  $1.23 \rightarrow 1.23 \rightarrow 1.13$ . Only **7d-6 strengthens**:  $1.52 \rightarrow 1.71 \rightarrow 1.93$ , and it is the one higher-risk cluster that remains significant like-for-like ( $q = 0.003$ ). Among the protective clusters, 7d-5 holds ( $0.50 \rightarrow 0.45 \rightarrow 0.52$ ,  $q = 0.03$  like-for-like) and 7d-3 holds in direction ( $0.57 \rightarrow 0.51 \rightarrow 0.44$ ) but loses significance like-for-like ( $q = 0.07$ ). The stable-dose clusters are all non-significant in this subset in both versions, including PCA-3 ( $1.34 \rightarrow 1.38 \rightarrow 1.29$ ,  $q = 0.06$  and  $0.38$ ).

The same picture holds for the continuous scalars (Figure 6). The post-test escalation odds ratio per SD is 1.28 pooled, 1.36 within-patient, 1.27 against observed-collection negatives with all positives, and **1.14** in the like-for-like comparison; attendance is 0.65, 0.58, 0.61 and 0.66 per SD across the same four analyses. Attendance is stable throughout; the escalation effect is directionally stable but its magnitude is roughly halved on the log scale in the like-for-like observed-collection comparison, mirroring the cluster-level result.

Two conclusions follow. First, the loss of significance for several higher-risk clusters is **not** attributable to reduced power alone: P/N is a ratio of occurrence rates and does not depend on sample size, and three of four higher-risk point estimates move toward the null. False-negative dilution is therefore not a general explanation of the full-negative-set results; it is at most a partial explanation for one cluster (7d-6). Second, this analysis does **not** rehabilitate PCA-3, and we do not offer it as doing so.

**Sensitivity to the imputation: curated versus raw-data reconstruction.** The escalation association was re-estimated on an independent reconstruction built directly from the source dosing spreadsheet. The two pipelines target the same estimand — the maximum three-day slope of the day-(-7)-normalised consumed dose over the post-test window — and differ in two respects: the curated pipeline (aim2.mat) computes the slope on the linearly interpolated series described under Inclusion and feature construction in the main text, whereas the raw reconstruction uses **only observed dosing days** and does not apply the window-completeness drop. On the curated features the association is OR **1.28** per SD ( $n = 9,900$ ,  $p = 3 \times 10^{-5}$ ); on the raw reconstruction it is OR **1.14** per SD ( $n = 8,560$ ,  $p = 3 \times 10^{-3}$ ). The association therefore replicates in direction and remains significant, but **roughly half of its magnitude on the log-odds scale depends on the interpolated days** rather than on observed doses. This is the same phenomenon as the imputation-free result under Sensitivity analyses in the main text, and it is a substantive limitation of the exposure definition rather than a reproducibility failure: both pipelines run, both agree in sign, and both are reported.

## **S2. Table S3b — Jaccard-matched bootstrap P/N intervals (dose-change branch)**

The 500-refit reproducibility run reported in Table S3 takes each refit's von Mises components at their raw index. Mixture components carry no fixed identity across refits, so an interval built that way compares different clusters and is inflated by label switching. The table below repeats the exercise with each refit's components first matched to the full-data clusters by maximum Jaccard overlap (greedy, without reuse). It is a separate analysis — 200 refits, 60 EM iterations per refit — run by 46\_audit\_bootstrap\_and\_ci.py, which writes matched\_bootstrap.json. That file, not final\_stability.json, is the source for the matched intervals below; final\_stability.json stores the 500-refit **unmatched** run reported in Table S3, and the two must not be conflated.

| Cluster           | Raw-index 95% interval | Width       | Jaccard-matched 95% interval | Width       |
|-------------------|------------------------|-------------|------------------------------|-------------|
| 7d-1              | 0.51–2.87              | 2.36        | 0.50–1.51                    | 1.01        |
| 7d-2              | 0.50–1.79              | 1.30        | <b>1.41–1.80</b>             | 0.40        |
| 7d-3              | 0.43–1.76              | 1.32        | 0.42–1.08                    | 0.66        |
| 7d-4              | 0.50–1.85              | 1.35        | <b>1.05–2.31</b>             | 1.26        |
| 7d-5              | 0.50–2.50              | 2.01        | 0.51–2.03                    | 1.51        |
| 7d-6              | 0.50–2.64              | 2.14        | 0.51–3.06                    | 2.56        |
| <b>mean width</b> |                        | <b>1.75</b> |                              | <b>1.23</b> |

Matching narrows the intervals by 29% on average. Two clusters then exclude unity (7d-2, 7d-4). This should be read alongside, not instead of, their behaviour in the other analyses: 7d-2 has no distinct outcome-blind counterpart and its within-patient contrast against 7d-1 is not significant, and 7d-4 is the single cluster that is not significant within-patient as a lone indicator (OR 1.12,  $p = 0.15$ ). Narrow resampling intervals therefore indicate that these two clusters are stably *located* under resampling, not that they are the best-supported clusters overall.

### S3. Table S6 — the alternative partition of the stable-dose branch

The stable-dose branch is reported with boundaries at the minima of a circular kernel-density estimate of the PCA angle. A one-dimensional von Mises mixture with  $K = 3$ , fitted to the same 1,336 stable-dose positive tests, is an equally defensible partition of the same data and gives a materially different answer. Both are shown below (48\_stabledose\_alt\_partition.py, which writes table\_S6\_alt\_partition.json).

| Partition                               | Cluster      | Positive n | Negative n | P/N         | q      | Label              |
|-----------------------------------------|--------------|------------|------------|-------------|--------|--------------------|
| Reported<br>(KDE<br>density<br>valleys) | PCA-1        | 908        | 2,240      | 0.77        | <0.001 | lower-risk         |
|                                         | PCA-2        | 161        | 471        | 0.65        | <0.001 | lower-risk         |
|                                         | <b>PCA-3</b> | 267        | 381        | <b>1.34</b> | <0.001 | <b>higher-risk</b> |
| Alternative<br>(von Mises,<br>$K = 3$ ) | 1            | 295        | 766        | 0.74        | <0.001 | lower-risk         |

|  |   |     |       |      |       |                 |
|--|---|-----|-------|------|-------|-----------------|
|  | 2 | 597 | 1,450 | 0.79 | 0.010 | lower-risk      |
|  | 3 | 444 | 876   | 0.97 | 0.622 | not significant |

The two partitions agree that part of the stable-dose branch is protective, and they disagree about whether any part of it is higher-risk: the reported partition yields one higher-risk cluster (PCA-3), the alternative yields none. Neither partition was pre-specified, and the reported one is the one that produces a higher-risk cluster. Because that choice cannot be shown to have been made blind to the outcome, the whole stable-dose branch is presented as exploratory and PCA-3 in particular is not claimed (Clustering, and Cluster associations with morphine-positive urine, in the main text). The dose-change branch, which carries the paper's confirmatory findings, is unaffected by this.

#### **S4. Table S5 — imputation of non-attended days**

The main text cites Table S5 for the full breakdown of the imputation; it is generated by 38\_units\_descriptives.py and summarised here. Interpolated observations account for 15.1% of  $\pm 7$ -day and 15.8% of  $\pm 28$ -day observations for positive tests, and 7.9% and 8.7% respectively for negative tests, so roughly one day in six of a positive test's window is filled rather than observed. The generated file also records the proportion of tests containing at least one interpolated day, from which the imputation-free subsets used in the main text are drawn (5,644 tests over  $\pm 7$  days and 3,737 over  $\pm 28$  days), together with how often the day-(-7) normalisation anchor was itself interpolated. Within the complete-pre-test-week subset described above, none of the 6,631 tests has an imputed anchor; the anchor is not similarly guaranteed for the sample as a whole, which is why the imputation-free subsets are defined from the pre-fill series rather than from attendance. Participant-flow detail (43\_linkage\_dedup\_checks.py). The 677 same-patient-day repeat urine records the pipeline retained span 486 patient-days and reduce to 246 excess records inside the

step-3 sample, only 24 of them discordant, so most are repeated identical entries and not second specimens. De-duplicating step 3 moves the positive rate from 0.345 to 0.352; the conservative duplication check on the cluster table is reported above.

## **S5. Estimates reported in full**

Compressing the main text to the journal's length limit moved the following estimates here. They are reported in full so that no figure quoted in an earlier draft is lost.

Taxonomy versus a single scalar (50\_lrt\_and\_cluster\_shape.py). Adding the nine clusters to a continuous slope improves fit against the claimed exposure as well as the disclaimed one:

likelihood-ratio  $\chi^2 = 328$  ( $p < 10^{-65}$ ) against the pre-test slope,  $\chi^2 = 261$  ( $p < 10^{-50}$ ) against the post-test slope, and  $\chi^2 = 254$  ( $p < 10^{-49}$ ) against both together, each on 8 degrees of freedom.

Within- versus between-patient estimates. Refitting with patient fixed effects, using the 451 patients who contributed both outcomes, leaves the associations undiminished on a common scale: post-test escalation moves from 1.28 to 1.36 per SD ( $p = 8 \times 10^{-19}$ ), pre-test escalation from 1.14 to 1.15 ( $p = 3 \times 10^{-5}$ ), and attendance from 0.84 to 0.80 per 0.1 increment ( $p = 10^{-62}$ ), the first value of each pair being the pooled and the second the within-patient estimate. Entering each cluster in turn, 8 of 9 were significant within-patient in the direction of Table 2 of the main text, the strongest being 7d-5 (OR 0.35,  $p = 5 \times 10^{-27}$ ) and the exception 7d-4 (OR 1.12,  $p = 0.15$ ).

Imputation-free subsets in full (42\_correct\_imputation\_free.py). Post-test escalation falls within patients to 1.06 per SD (95% CI 0.97–1.15,  $p = 0.23$ ) on imputation-free  $\pm 7$ -day windows and rises to 1.44 (95% CI 1.30–1.60,  $p = 1 \times 10^{-12}$ ) on the complement. Pre-test escalation rises to 1.26 per SD (95% CI 1.15–1.38,  $p = 3 \times 10^{-7}$ ) over  $\pm 7$  days and to 1.37 (95% CI 1.20–1.56) over  $\pm 28$  days, against an inconclusive 1.07 ( $p = 0.16$ ) on the complement. Entering the clusters jointly with 7d-1 as reference, 7 of 8 contrasts remain significant on the imputation-free subset (n

253 = 4,280 of the 5,644 imputation-free tests, conditional logistic regression using only patients who  
254 contribute both a positive and a negative test, so that 1,364 tests from concordant patients drop  
255 out; strongest 7d-5, OR 0.14 against 7d-1,  $p = 5 \times 10^{-26}$ ). Those contrasts are against a reference  
256 cluster and are therefore on a different scale from the single-indicator estimates above, so the  
257 change reflects the model and not the restriction.

258 Confounding by indication (49\_confound\_same\_exposure.py). The prescribed-dose comparator  
259 is computed with the same algorithm as the exposure, namely the maximum three-day moving-  
260 window slope over the pre-test days. Adjusting the consumed-dose association for it moves the  
261 within-patient odds ratio from 1.153 to 1.144 per SD on all tests, and from 1.259 to 1.236 (95%  
262 CI 1.13–1.35,  $p = 2 \times 10^{-6}$ ) on imputation-free  $\pm 7$ -day windows, the first value of each pair being  
263 the unadjusted estimate. The prescribed slope taken alone carries no clear signal on the full  
264 sample (OR 1.03,  $p = 0.27$ ) but is associated with the result on imputation-free windows (OR  
265 1.13, 95% CI 1.04–1.22,  $p = 0.003$ ), independently of consumption.

266 Incremental predictive value (33\_max\_predict.py, 20\_pretest\_only.py). Under patient-grouped  
267 cross-validation a baseline of pre-test attendance plus prior urine-test history reached an area  
268 under the curve of approximately 0.83; adding the full pre-test dose-trajectory features changed  
269 this by approximately 0.00. Dose-trajectory features restricted to strictly pre-test data reached  
270 approximately 0.60, and attendance alone approximately 0.58. An exhaustive search over richer  
271 feature sets, personalised baselines, stratification and change-point formulations (scripts 28 to 33)  
272 did not raise the ceiling.